Week 2 Deliverable: Regolith Transport and Payload Trade Calculations

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1 Week 2 Objective

The objective of Week 2 is to connect the lunar regolith mover rover concept to the previous lunar radiation shielding project. The main goal is to estimate how much regolith must be transported for the medium and heavy shielding cases, then determine how many rover trips would be required for different payload capacities and fleet sizes.

This week focuses on the following questions:

1. How much regolith must be moved for the medium and heavy shielding configurations?
2. What rover payload sizes should be considered?
3. How many total trips are required for each payload case?
4. How does the required trip count change if a fleet of 10–15 rovers is used?
5. How long would the regolith transport operation take under simple route assumptions?

2 Connection to Radiation Shielding Project

The previous lunar habitat shielding project studied regolith-based shielding for a habitat structure. The rover project extends that work by addressing the practical deployment question: how the required regolith shielding material could be transported and placed around the habitat.

For this analysis, two shielding cases are considered:

- **Medium shield:** 0.75 m regolith shielding layer
- **Heavy shield:** 1.50 m regolith shielding layer

The total regolith masses from the previous shielding analysis are:

$$M_{medium} = 61,200 \text{ kg}$$

$$M_{heavy} = 139,000 \text{ kg}$$

These values represent the total amount of regolith that must be transported by the rover system to construct the shielding configuration.

3 Payload Cases

Instead of choosing only one rover payload, three payload cases are analyzed. This allows the project to compare smaller rover designs with larger heavy-hauler concepts.

Rover Class	**Payload Mass**	**Earth-lb Equivalent**
Medium Rover	100 kg	220 lb
Large Rover	250 kg	551 lb
Heavy Hauler	450 kg	992 lb

Table 1: Payload cases considered for the Week 2 regolith transport analysis.

The 100 kg case represents a conservative medium rover. The 250 kg case represents a larger 500 lb-class rover. The 450 kg case represents a 1000 lb-class robotic lunar dump truck or heavy regolith hauler.

For this project, the 250 kg case will be treated as the safer large-rover case, while the 450 kg case will be treated as an ambitious heavy-hauler case.

4 Trip Count Equation

The number of rover trips required is calculated using:

$$N_{trips} = \frac{M_{regolith,total}}{m_{payload}}$$

where:

- N_{trips} = total number of rover trips required
- $M_{regolith,total}$ = total regolith mass required for shielding
- $m_{payload}$ = regolith mass carried per rover trip

5 Total Trips Required

5.1 Medium Shield Case

For the medium shield:

$$M_{medium} = 61,200 \text{ kg}$$

Payload Case	Payload Mass	Total Trips Required
Medium Rover	100 kg	612 trips
Large Rover	250 kg	245 trips
Heavy Hauler	450 kg	136 trips

Table 2: Total trips required to transport regolith for the medium shielding case.

5.2 Heavy Shield Case

For the heavy shield:

$$M_{heavy} = 139,000 \text{ kg}$$

Payload Case	Payload Mass	Total Trips Required
Medium Rover	100 kg	1390 trips
Large Rover	250 kg	556 trips
Heavy Hauler	450 kg	309 trips

Table 3: Total trips required to transport regolith for the heavy shielding case.

6 Fleet-Based Rover Operation

A single rover would require a very large number of trips, especially for the heavy shielding case. Therefore, this project assumes that the regolith transport operation would use a fleet of multiple rovers working together.

The fleet sizes considered are:

$$n_{rovers} = 10,\ 12,\ 15$$

The number of trips per rover is:

$$N_{per\ rover} = \frac{N_{trips}}{n_{rovers}}$$

7 Trips Per Rover for Medium Shield

Payload Case	10 Rovers	12 Rovers	15 Rovers
100 kg	62 trips/rover	51 trips/rover	41 trips/rover
250 kg	25 trips/rover	21 trips/rover	17 trips/rover
450 kg	14 trips/rover	12 trips/rover	10 trips/rover

Table 4: Trips per rover for the medium shielding case using different fleet sizes.

8 Trips Per Rover for Heavy Shield

Payload Case	10 Rovers	12 Rovers	15 Rovers
100 kg	139 trips/rover	116 trips/rover	93 trips/rover
250 kg	56 trips/rover	47 trips/rover	38 trips/rover
450 kg	31 trips/rover	26 trips/rover	21 trips/rover

Table 5: Trips per rover for the heavy shielding case using different fleet sizes.

9 Route Distance Assumptions

The rover will transport regolith from a collection zone to a habitat shielding zone. Since the exact base layout has not yet been finalized, three one-way route distances are considered:

$$d_{one-way} = 25\text{ m},\ 50\text{ m},\ 100\text{ m}$$

The round-trip distance is:

$$d_{round} = 2d_{one-way}$$

One-Way Distance	Round-Trip Distance
25 m	50 m
50 m	100 m
100 m	200 m

Table 6: Possible rover route distance assumptions.

10 Trip Time Calculation

The time required for one complete trip includes driving time, loading time, and dumping time.

$$t_{trip} = t_{drive} + t_{load} + t_{dump}$$

The driving time is:

$$t_{drive} = \frac{d_{round}}{v}$$

where:

- t_{drive} = driving time
- d_{round} = round-trip distance
- v = rover driving speed

For this preliminary analysis, the rover speed is assumed to be:

$$v = 0.25 \text{ m/s}$$

The loading and dumping times are assumed to be:

$$t_{load} = 5 \text{ min}$$

$$t_{dump} = 2 \text{ min}$$

10.1 Example Trip Time

For a 50 m one-way route:

$$d_{round} = 100 \text{ m}$$

$$t_{drive} = \frac{100}{0.25} = 400 \text{ s}$$

$$t_{drive} = 6.67 \text{ min}$$

Therefore:

$$t_{trip} = 6.67 + 5 + 2$$

$$t_{trip} = 13.67 \text{ min}$$

For simplicity:

$$t_{trip} \approx 14 \text{ min}$$

11 Fleet Operation Time

The ideal fleet operation time is:

$$t_{fleet} = \frac{N_{trips} t_{trip}}{n_{rovers}}$$

where:

- t_{fleet} = ideal total fleet operation time
- N_{trips} = total trips required
- t_{trip} = time per trip
- n_{rovers} = number of rovers in the fleet

Because real rover operations will not be perfectly efficient, an efficiency factor is included:

$$\eta = 0.7$$

The realistic operation time is:

$$t_{real} = \frac{t_{fleet}}{\eta}$$

12 Example Operation Time: Heavy Shield, Heavy Hauler Case

For the heavy shield with 450 kg payload rovers:

$$N_{trips} = 309$$

Assuming:

$$t_{trip} = 14 \text{ min}$$

$$n_{rovers} = 15$$

The ideal fleet time is:

$$t_{fleet} = \frac{309(14)}{15}$$

$$t_{fleet} = 288.4 \text{ min}$$

$$t_{fleet} = 4.81 \text{ hr}$$

Including the efficiency factor:

$$t_{real} = \frac{4.81}{0.7}$$

$$t_{real} = 6.87 \text{ hr}$$

Therefore, under these assumptions, a fleet of 15 heavy-hauler rovers could transport the regolith required for the heavy shield in approximately:

$$\boxed{6.9 \text{ hr}}$$

This value is an idealized preliminary estimate and does not include battery recharge time, maintenance, terrain delays, traffic management, or loading-system limitations.

13 Week 2 Discussion

The trip calculations show that rover payload capacity has a major effect on the number of required trips. A 100 kg rover is easier to design but requires a much larger number of trips. A 250 kg rover provides a reasonable large-rover baseline and reduces the trip count significantly. A 450 kg rover, representing a 1000 lb-class robotic lunar dump truck, greatly reduces the total number of trips but would require a larger chassis, stronger motors, better suspension, wider wheelbase, and more careful center-of-gravity control.

Using multiple rovers also makes the construction concept more realistic. Instead of relying on a single rover to complete hundreds or thousands of trips, a fleet of 10–15 rovers can divide the workload. This reduces the number of trips per rover and makes the construction schedule more practical.

The 450 kg heavy-hauler case appears attractive from an operations standpoint because it reduces the heavy shield case to approximately 21–31 trips per rover for a 10–15 rover fleet. However, this case should be analyzed carefully in later weeks for wheel torque, power demand, stability, and structural loading.

14 Week 2 Conclusion

The Week 2 analysis connects the rover concept directly to the lunar habitat radiation shielding project. The medium shielding case requires approximately:

$$\boxed{61,200 \text{ kg of regolith}}$$

while the heavy shielding case requires approximately:

$$\boxed{139,000 \text{ kg of regolith}}$$

Three rover payload cases were considered: 100 kg, 250 kg, and 450 kg. The 250 kg case is a reasonable large-rover baseline, while the 450 kg case represents a 1000 lb-class heavy-hauler rover. For a fleet of 10–15 rovers, the heavy-hauler case reduces the required trips per rover to approximately 21–31 trips for the heavy shielding case.

Based on this analysis, the project will continue with the following primary design assumptions:

$$\boxed{m_{payload} = 250\text{–}450 \text{ kg per rover}}$$

$$\boxed{n_{rovers} = 10\text{–}15}$$

$$\boxed{v = 0.25 \text{ m/s}}$$

$$\boxed{t_{trip} \approx 14 \text{ min for a 50 m one-way route}}$$

The next step is to use these assumptions to begin defining the rover mission architecture, subsystem breakdown, and basic vehicle design.